The Output Side Algorithm

The first viewpoint of convolution analyzes how each sample in the *input signal* affects many samples in the output signal. In this second viewpoint, we reverse this by looking at individual samples in the *output signal*, and finding the contributing points from the input. This is important from both mathematical and practical standpoints. Suppose that we are given some input signal and impulse response, and want to find the convolution of the two. The most straightforward method would be to write a program that loops through the *output signal*, calculating one sample on each loop cycle. Likewise, equations are written in the form: $y[n] = \textit{some combination of other variables}$. That is, sample n in the output signal is equal to some combination of the many values in the input signal and impulse response. This requires a knowledge of how each sample in the output signal can be calculated independently of all other samples in the output signal. The output side algorithm provides this information.

Let's look at an example of how a single point in the output signal is influenced by several points from the input. The example point we will use is $y[6]$ in Fig. 6-5. This point is equal to the sum of all the sixth points in the nine output components, shown in Fig. 6-6. Now, look closely at these nine output components and identify which can affect $y[6]$. That is, find which of these nine signals contains a nonzero sample at the sixth position. Five of the output components only have *added* zeros (the diamond markers) at the sixth sample, and can therefore be ignored. Only four of the output components are capable of having a nonzero value in the sixth position. These are the output components generated from the input samples: $x[3]$, $x[4]$, $x[5]$, and $x[6]$. By adding the sixth sample from each of these output components, $y[6]$ is determined as: $y[6] = x[3]h[3] + x[4]h[2] + x[5]h[1] + x[6]h[0]$. That is, four samples from the input signal are multiplied by the four samples in the impulse response, and the products added.

Figure 6-8 illustrates the output side algorithm as a **convolution machine**, a flow diagram of how convolution occurs. Think of the input signal, $x[n]$, and the output signal, $y[n]$, as fixed on the page. The convolution machine, everything inside the dashed box, is free to move left and right as needed. The convolution machine is positioned so that its output is aligned with the output sample being calculated. Four samples from the input signal fall into the inputs of the convolution machine. These values are multiplied by the indicated samples in the impulse response, and the products are added. This produces the value for the output signal, which drops into its proper place. For example, $y[6]$ is shown being calculated from the four input samples: $x[3]$, $x[4]$, $x[5]$, and $x[6]$.

To calculate $y[7]$, the convolution machine moves one sample to the right. This results in another four samples entering the machine, $x[4]$ through $x[7]$, and the value for $y[7]$ dropping into the proper place. This process is repeated for all points in the output signal needing to be calculated.